

FAQs – EFS-02

1. What confidence level should we use to check for stationarity?

Practitioners generally use 95% and 99% confidence levels to check for stationarity in a time series.

2. If most financial assets are non-stationary, what can we do?

Financial asset price movements are mostly non-stationary while their returns follow a stationary process. We can either

- work with the returns series for analytical tractability or
- use multiple assets (two or more) to create a stationary spread using those assets. Then the spread can be traded.
- Consider log transformation of data

3. How is normal arbitrage different from statistical arbitrage?

In normal arbitrage, traders try to earn from the discrepancies in the prices in the same asset on two or more trading venues. For example, if *TCS* trades at price X on NSE and at price $X+1$ on the BSE, we can go long on the NSE and short on the BSE to lock a riskless profit of 1.

On the other hand, statistical arbitrage is a type of arbitrage where traders use statistical properties and try to identify the discrepancies between two or more assets and trade them.

4. Should we account for the difference in overnight interest when trading forex pairs?

We should ideally account for it along with commissions and slippages for close-to-real backtesting.

5. What is a hedge ratio?

For statistical arbitrage, the hedge ratio tells us how much quantity of one stock is required to hedge another stock. For example, how many shares of stock X do we need to go long when we short stock Y .

6. For implementing statistical arbitrage, what do we need to check? Co-integration or correlation?

Before we can trade stocks, we need to ensure that the spread between them is stationary. Hence, we need to check for the co-integration of the spread.

7. For triplets trading, how to identify the hedge ratio for each?

We can compute the hedge ratio of each stock by regressing it against the target stock. We demonstrate this idea in the assignment.